

# Requirements and service boundaries

Project: Grounded Learning Service

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Neural network experiment defense

Without opening the old lesson, explain one key idea from Neural network experiment defense. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Trace the worked case for Requirements and service boundaries. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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### 3. Individual reasoning check

Use the worked case for Requirements and service boundaries. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

#### **My explanation before partner discussion**

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### 4. Independent transfer case

#### **Independent case: Apply a new case independently**

A read-only inventory service is asked to list item A and delete item B. Inventory says A is available. Trace permitted work.

#### **My independent explanation, trace and limits**

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### 5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

**My reasoning and evidence**

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## 6. W2 Trace the worked case

A request asks for Tuesday meeting time and to move it to Wednesday. The public schedule lists Tuesday 16:00. Show the intermediate steps and give the conclusion with its scope.

**My reasoning and evidence**

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## 7. W3 Audit the limit

A report omits this qualification: A helpful answer does not grant an operation that the service contract excludes. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**My reasoning and evidence**

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## 8. W4 Apply a new case independently

A read-only inventory service is asked to list item A and delete item B. Inventory says A is available. Trace permitted work.

### My reasoning and evidence

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## 9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

### My reasoning and evidence

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## 10. Project checkpoint and reflection

Add a labelled requirements and service boundaries evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of requirements and service boundaries. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

### My project change, supporting evidence and remaining limit

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## 11. Feedback, return check and learning record

### Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_  
\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 12. Pictorial case practice: Requirements and service boundaries

Museum fact desk. A visitor asks the opening year of exhibit P and asks the bot to sell a ticket.

Approved card P says 1912. The contract permits reading cards, not selling tickets.

AI ACADEMY / AI-6/1

### Requirements and service boundaries

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



**1. Read approved card**

The supplied year is 1912.



**2. Split the request**

Answering the year and selling a ticket are separate operations.



**3. Apply the boundary**

Return 1912 with card P; do not sell a ticket.

**CHANGE ONE THING**

Replace P with Q, which has no approved card.

Explain the changed result using the matching lesson.

Split the museum visitor's request into two operations. Which can the service perform for P, and what happens to each operation when Q has no approved card?

### My picture-based explanation and changed-case prediction

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

## 13. Project portfolio milestone

Project: Grounded Learning Service

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

### Evidence to keep · My record

**Evidence to keep:** This week's input and deciding step

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** Prediction and actual result (or labelled paper trace)

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** One change since the previous checkpoint and why

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** One error or limit and the next test

**My record:** \_\_\_\_\_

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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## 14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Museum fact desk. A visitor asks the opening year of exhibit P and asks the bot to sell a ticket.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Return 1912 with card P; do not sell a ticket.

1. Read approved card
2. Split the request
3. Apply the boundary

**My selected conclusion, deciding fact and explanation**

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Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Apply the boundary Return 1912 with card P; do not sell a ticket. A service boundary states permitted work; useful information does not grant extra operations.

## Practice 2: explain the deciding evidence

Changed-case practice: Replace P with Q, which has no approved card. Which conclusion is supported by the supplied facts?

1. The year is unsupported, and the ticket sale is still excluded.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

### My selected conclusion, deciding fact and explanation

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Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The year is unsupported, and the ticket sale is still excluded. Knowledge and permission are separate requirements.

## Repair a missing building block

Review Neural network experiment defense (ai-5/36). Goal: Defend a network experiment using traceable evidence. Short practice: Calculate both weighted terms,  $z$  and ReLU output. Predict the changed second input and explain why 0.2 cannot be called 20% accuracy.. Return to this week and explain what you changed.

## Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

### What I repaired, the evidence I kept and the next test

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